

â€” The correction is $\sim 10 \hat{A}^2 \hat{A}^3$ at 1 AU â€” negligibly small in the Solar System but significant at stellar wind compressed regions ($r \ll 1$ AU, $\tilde{\rho} \gg \tilde{\rho}_{sw}(1 \text{ AU})$).

3. Corrected Ubi Equation

$$U_{b,i}(r, t) = -\beta_i \cdot U_{gi}(r, t) \cdot \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot wind_mod \cdot U_{UA} \cdot \cos(\pi t_n)$$

where $wind_mod = 1 + \epsilon_{sw} \cdot \rho_{sw}(r)$ and $\tilde{\rho}_{sw}(r)$ falls off as:

$$\rho_{sw}(r) = \rho_{sw,0} \cdot \left(\frac{r_0}{r}\right)^2$$

at $r_0 = 1$ AU (density decreases as inverse square with distance).

4. Extended Buoyancy â€” H_SCm Integration

The superconductive medium height H_{SCm} (PAPER_064 canonical H_SCm â€” 0.99) also enters the buoyancy via:

$$U_{b,i} \propto H_{SCm} \cdot wind_mod$$

For $H_{SCm} = 0.99$ (99% SCm coverage):

$$U_{b,i}(H_{SCm}) = -\beta_i \cdot U_{gi} \cdot \Omega_g \cdot M_{bh}/d_g \cdot H_{SCm} \cdot wind_mod \cdot U_{UA} \cdot \cos(\pi t_n)$$

5. Physical Mechanism

The solar wind density modulates the buoyancy force through three physical channels:

1. **Plasma density effect:** Higher $\tilde{\rho}_{sw}$ increases the ambient medium density, which increases the buoyant uplift (Archimedes principle: $F_b \hat{=} \tilde{\rho}_{fluid} \tilde{V} \tilde{g}$)
 2. **Ram pressure effect:** Solar wind ram pressure $P_{ram} = \tilde{\rho}_{sw} v_{sw}^2/2$ compresses the magnetosphere boundary, altering the effective R_b and thus U_g
 3. **Ion-neutral friction:** Wind ions interact with neutral UQFF vacuum, transferring momentum â€” drift term in Ubi
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6. Solar Wind Density at Different Radii

Location	r/AU	$\tilde{\rho}_{sw}$ [kg/m ³]	wind_mod
Solar corona	0.01	8.35×10^{-12}	$1 + 8.35 \times 10^{-12} \times \tilde{\rho}_{sw}^0$
Mercury	0.39	5.48×10^{-12}	$1 + 5.48 \times 10^{-12} \times \tilde{\rho}_{sw}^2$
Earth (1 AU)	1.0	8.35×10^{-12}	$1 + 8.35 \times 10^{-12} \times \tilde{\rho}_{sw}^1$
Jupiter (5 AU)	5.2	3.09×10^{-12}	$1 + 3.09 \times 10^{-12} \times \tilde{\rho}_{sw}^2$
Termination	100	8.35×10^{-12}	$1 + 8.35 \times 10^{-12} \times \tilde{\rho}_{sw}^2$

The wind_mod correction only becomes dynamically significant (>1%) at $\tilde{\rho}_{sw} > 10^3$ kg/m³, which corresponds to conditions inside accretion disks or dense stellar winds.

7. Consistency with Ug2 ($\hat{\rho}'_{sw}$ Parameter)

The Ug2 term used $\hat{\rho}'_{sw} \cdot v_{sw}$ with $\hat{\rho}'_{sw} = 0.001$ and $v_{sw} = 400$ km/s:

$$1 + \delta_{sw} v_{sw} = 1 + 0.001 \times 4 \times 10^5 = 1.4 \quad (40\% \text{ correction})$$

The new wind_mod is consistent when $\tilde{\rho}_{sw}$ is calibrated as an **equivalent pressure**:

$$\epsilon_{sw} \cdot \rho_{sw} \equiv \delta_{sw} \cdot v_{sw}$$

$\tilde{\rho}_{sw} = \hat{\rho}'_{sw} \cdot v_{sw} / \hat{\rho}_{sw} = 0.001 \times 4 \times 10^5 / 0.001 = 4 \times 10^8 \text{ kg/m}^3$
(accretion disk density)

This confirms $\hat{\rho}_{sw} = 0.001$ as the correct coupling for dense stellar wind environments.

8. CP Integration

CP2 update: Add $\epsilon_{sw} = 0.001$ to UQFIBuoyancyCalculator. Implement `compute_wind_mod(rho_sw, epsilon_sw=0.001)` and apply to all Ubi terms.

Status: Complete | **CP Stage:** CP2 **Supersedes:** N/A (clarifies $\hat{\rho}'_{sw}$ vs $\hat{\rho}_{sw}$)
| **Related:** PAPER_064 (4 operational modes, H_SCm), PAPER_086 (Ubi derivation), PAPER_157 (Solar System Ubi)

UQFF computed: Solar wind UQFF correction = $[SSq] \times \exp(-? \times r/v) = 5.7e-1 \times \exp(-5.0e-4 \times (1\text{AU}/400\text{km/s})) = 5.7e-1 \times \exp(-3.2e-3) \approx 5.7e-1$; dominant at $r < 1\text{AU}$.